Vuong Hung Truong

thvuong234@gmail.com vuonghtruong

Research Interests

Circadian rhythms; T-cycle; Arduino Programing; Non-linear dynamics; Limit Cycle; Cognitive Science; Machine Learning; Reinforcement Learning; Large Language Models; Neuroimaging (fMRI, EEG); Open Science

Current Position

Taipei Medical University, PI: Dr. Jihwan Myung **2024 – now**
PhD student

Project involved investigating the circadian rhythms under T-cycles, combining both empirical data (bioluminescence) and entrainment theory.

Education

Eberhard Karls Universität Tübingen, PI: Dr. Andreas Bartels **2020 – 2023**
International Max Planck Research Schools' PhD program in Neural & Behavioural Sciences

Project involved using Volatile Kalman Filter to quantify environmental volatility in reversal learning and correlating that with fMRI signals.

Taipei Medical University, PI: Dr. Niall Duncan **2017 – 2019**
Master's program in Cogntion and Consciousness *GPA: 4/4*
Project involved using Hierarchical Gaussian Filter to generate prediction error trajectories and correlating that with EEG signals.

School of Medicine - Vietnam National University, PI: Dr. Cam Huyen Nguyen Xuan **2010 – 2016**
Doctor of Medicine *GPA: 3.5/4*
Project involved descriptive statistics of Guillain–Barré syndrome.

Programming experience

Python

Machine learning, deep learning, LLM, cognitive models (e.g. Rescorla–Wagner, Kalman Filter, Hidden Markov Model, etc...)

C++

Programming micro-controllers to control LED and sensors, analyzing EEG data

Mathematica

Analyzing non-stationary time-series and bio-imaging data

Awards

Third position at TMU-MIT (Sana) Hackathon on Health IoT **October 2018**
Taipei Medical University & Massachusetts Institute of Technology

Award for the 1st prize in English Eloquence Competition **2014**
HCMC University of Medicine and Pharmacology

Award for the highest score in the 1st Academic year **2011**
Vietnam National University

Publications

Chang, P., Hitrec, T., Muir, C., Sofyana, M., **Truong, V. H.**, Lacey, S., ... & Piggins, H. D. (2026). Differential effects of sex and age on daily and infradian rhythms of mice. *The Journal of Physiology*.

Binz, M., Akata, E., Bethge, M., Brändle, F., Callaway, F., Coda-Forno, J., ...(including **Truong, V. H.**) & Schulz, E. (2025). A foundation model to predict and capture human cognition. *Nature*, 644(8078), 1002–1009.

Myung, J., Vitet, H., **Truong, V. H.**, & Ananthasubramaniam, B. (2025). The role of the multiplicity of circadian clocks in mammalian systems. *Sleep Medicine*, 131, 106518.

Mugleston, J.*, **Truong, V. H.***, Kuang, C., Sibiya, L., & Myung, J. (2025). Epistemology in the age of large language models. *Knowledge*, 5(1), 3.(* Equal contribution)

Truong, V., & Myung, J. (2023). LocoBox: Modular Hardware and Open-Source Software for Circadian Entrainment and Behavioral Monitoring in Home Cages. *Sensors*, 23(23), 9469.

Rawal, A.*, **Truong, V.***., Lo, Y. H., Tseng, L. Y., & Duncan, N. W. (2024). A survey of experimental stimulus presentation code sharing in major areas of psychology. *Behavior Research Methods*, 1–11. (* Equal contribution)

Truong, V., Cheng, P.Z., Lee, H.C., Lane, T.J., Hsu, T.Y. and Duncan, N.W., 2021. Occipital gamma-aminobutyric acid and glutamate-glutamine alterations in major depressive disorder: An MRS study and meta-analysis. *Psychiatry Research: Neuroimaging*, 308, p.111238.

Truong, V. and Duncan, N.W., 2020. Suggestions for improving the visualization of magnetic resonance spectroscopy voxels and spectra. *Royal Society Open Science*, 7(8), p.200600.

Botvinik-Nezer, R., Holzmeister, F., Camerer, C.F. [et al, including **Truong, V.**], 2020. Variability in the analysis of a single neuroimaging dataset by many teams. *Nature*, 582(7810), pp.84–88.

Myung, J., Wu, M.Y., Lee, C.Y., Rahim, A.R., **Truong, V.**, Wu, D., Piggins, H.D. and Wu, M.S., 2019. The kidney clock contributes to timekeeping by the master circadian clock. *International Journal of Molecular Sciences*, 20(11), p.2765.

Courses taken (Graduate)

Computational Cognitive Science - University of Tübingen **2022**
Learning about computational modeling of human behaviors from model definition, fitting and validation.

Essential Mathematics for Neuroscience - University of Tübingen **2020–2021**
Learning about Calculus, Linear Algebra, and Advanced Topics such as PCA and Fourier analysis.

Mathematical Methods in Neuroscience - Taipei Medical University **2018–2019**
Learning about Ordinary Differential Equations and basic concepts in modeling the circadian system.

Teaching

Experimental Cognitve Science - University of Tübingen **2022–2023, 2021–2022**
Lecturer – involved in teaching, programming online behavioral experiments with PsychoPy & HTML

Supervision

Jinyang Zheng (Neil), Bachelor, Taipei Medical University **2024**
Meta-analysis of factors influencing behavioral circadian rhythm's period in mice.

Sahiti Chebolu, MSc, Max Planck Institute for Biological Cybernetics, Tübingen **2023**
Investigating the difference between monetary rewards and visual rewards in a two-armed bandit task.

Weichen Bai, Bachelor, Taipei Medical University **2020**
Building a mock MRI scanner with stereoscopic camera for measuring head motion and physiological signals.